

YEONGBIN KWON

EDUCATION

Columbia University

New York, NY, United States

Master of Science in Civil Engineering and Engineering Mechanics

2026.08 - 2027.12 (expected)

Seoul National University

Seoul, Republic of Korea

Bachelor of Science in Civil and Environmental Engineering

2022.03 - 2026.02

- Graduated with honors: *summa cum laude*
- Received Outstanding Undergraduate Student Commendation by the Korean Society of Coastal Disaster Prevention
- Graduated with Bachelor's thesis *Calibration-Free Stereo Imaging for Coastal Wave Field Reconstruction and Quantification of Wave Breaking*. Selected as the Best Bachelor's Thesis of Fall 2025.

Sejong Science High School

Seoul, Republic of Korea

Early Graduate — Graduated a year earlier than the usual three years program 2020.03 - 2022.02

PUBLICATIONS

1. Qin Huang, Moyan Liu, **Yeongbin Kwon**, Sarah Kapnick, and Upmanu Lall. Steering Tropical Cyclones Through Targeted Thermodynamic Perturbations in an AI Weather Model. *JGR: Machine Learning and Computation*, 2026. (under review)
2. Qin Huang, Moyan Liu, **Yeongbin Kwon**, and Upmanu Lall. Evaluating the Predictability of Selected Weather Extremes with Aurora, an AI Weather Forecast Model. *Atmosphere*, 17(8):716, 2026. doi:10.3390/atmos17080716

POSTERS

AND

PRESENTATIONS

1. **Yeongbin Kwon**, Hyoseob Noh, and Yong Sung Park. Calibration-Free Stereo Imaging for Coastal Wave Field Reconstruction and Quantification of Wave Breaking. *IAHR-APD 2026*, 2026. (poster presentation)
2. Moyan Liu, Qin Huang, Upmanu Lall, **Yeongbin Kwon**, and Hyun-Han Kwon. Weather Jiu-Jitsu: Toward Controlling Atmospheric Rivers with Deep Learning Foundation Models and Adaptive Perturbation. *AGU25 Annual Meeting*, 2025. (poster presentation)
3. Qin Huang, Moyan Liu, Upmanu Lall, **Yeongbin Kwon**, Hyun-Han Kwon, and Yuan Liu. Weather Jiu-Jitsu: Steering Tropical Cyclones with Small Perturbations of the Jet Stream in Experiments with Deep Learning Foundation Models. *AGU25 Annual Meeting*, 2025. (poster presentation)
4. Upmanu Lall, Qin Huang, Moyan Liu, Hyun-Han Kwon, and **Yeongbin Kwon**. Weather Jiu Jitsu: Adaptive Chaos Control with Deep Learning of Atmospheric Dynamics to leverage the power of the atmospheric circulation with nudges to move or defuse storms and avert catastrophic flooding. *AGU25 Annual Meeting*, 2025. (Presented by Upmanu Lall)
5. Upmanu Lall, Moyan Liu, Qin Huang, Hyun-Han Kwon, Adam Nayak, **Yeongbin Kwon**, and Mengjie Zhang. Weather Jiu Jitsu: A 21st century paradigm to leverage nature's power to mitigate some Compound, Concurrent, and Cascading Weather Hazards in a Changing Climate. *AGU25 Annual Meeting*, 2025. (Presented by Upmanu Lall)

PROJECTS	Calibration-Free Stereo Imaging for Coastal Wave Field Reconstruction and Quantification of Wave Breaking	SNU College Undergraduate Independent Research Grant	2025.05 - 2025.12
	Using stereo imagery to reconstruct wave fields and analyze wave breaking phenomena in real coastal environments.		
INTERNSHIPS AND EXPERIENCES	Low Impact Development (LID) Strategies for Stormwater Reduction and Reuse on Campus	SNU Integrated Design of Civil Engineering Systems Course	2025.09-2025.12
	- Evaluating and comparing the performance of various LID strategies at the SNU campus, using SWMM simulations based on varied inputs. - Selected as Outstanding Graduation Project (Top 3).		
	Researcher Institute of Engineering Research, SNU		2026.03 - 2026.08
	Conducted post-graduation research at CELAB on identifying rip currents in coastal environments.		
	Undergraduate Research Intern Columbia University, New York City		2025.06 - 2025.08
COMMUNITY SERVICE	- Used AURORA, one of the most recent foundational models in climate, to explore weather dynamics and how perturbations affects the dynamics. - Worked with Prof. Upmanu Lall's group in the Department of Earth and Environmental Engineering; contributed two AGU25 conference abstracts, a first-author-team paper published in <i>Atmosphere</i>, and a follow-up manuscript on cyclone steering now under review at <i>JGR: Machine Learning and Computation</i>.		
	Trilateral Students Activity in Seoul Seoul National University, Republic of Korea		2024.08
	Participated in a trilateral seminar between SNU, National Taiwan University, and University of Tokyo as the Geotechnical Engineering team leader, leading discussions.		
	Visiting Student University of California, Berkeley		2024.06 - 2024.08
ACADEMIC HONORS AND AWARDS	- Selected as a recipient of SNU Study Abroad Program grant. - Took classes including <i>Introduction to Partial Differential Equations</i> at UC Berkeley; lived in the International House and participated in diverse cultural activities.		
	Undergraduate Mentor Seoul National University Social Responsibility		2025.05 - 2025.12
	Mentoring children in middle school, focusing on interaction and communication with the student.		
	Undergraduate Volunteer Seoul National University Social Responsibility		2025.03 - 2025.06
	- Mentored a middle school student in Korea from a Bolivian-Korean multi-cultural family for three months. Helped the student based on cultural interactions, especially focusing on education. - Outstanding Volunteer Award recipient (2025.07)		
	Study Abroad Scholarship (25th Cohort) Kwanjeong Foundation		2026
	Outstanding Undergraduate Student Commendation Korean Society of Coastal Disaster Prevention		2026.02
	Best Bachelor's Thesis Award SNU		2025.12
	Outstanding Graduation Project Award SNU		2025.12
	Merit-Based Tuition Scholarship SNU		Fall 2025
	Outstanding Volunteer Award Seoul National University Social Responsibility		2025.07
	Short-Term Study Abroad Scholarship SNU Office of International Affairs		Spring 2024
	Merit-Based Tuition Scholarship SNU		Spring 2024
	Gwanak Scholarship (Merit-Based) Gwanak Society		Fall 2023

SKILLS

Languages: Korean (native), English (native), Japanese (proficient and fluent).

Programming and Typesetting: $\text{\LaTeX}$, Python, MATLAB, R, C, HTML/CSS, Deep Learning.